

# IMPLICIT-FD: Can a Full-Duplex Speech Model Infer the Turn-Taking Policy a Persona Implies?

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## Abstract

Deployed full-duplex (FD) voice agents are almost never configured with an explicit turn-taking directive; they are configured with a *role*. Nobody writes “interrupt when the user states a factual error”—they write “you are a Socratic math tutor,” and the floor-management policy is left *entailed* by the role. We introduce IMPLICIT-FD, an evaluation-only benchmark that asks a strictly harder question than prior work: can a full-duplex model reason from a persona to the turn-taking policy that persona implies, and then execute that policy in real time? IMPLICIT-FD instantiates every conversation at five conditioning levels—bare (L0), explicit directive (L1), persona-only (L2), persona-plus-restated-rule (L3), and persona-in-conflict-with-a-directive (L4)—over 8 assistant roles, 6 probes, and 5 instances, yielding 1038 test cases from 240 pre-synthesised two-channel conversations. Our headline metric, the ENTAILMENT\_GAP =  $IAS(L1) - IAS(L2)$ , isolates the cost of *having to infer* the rule. Scoring is kept on two strictly separated spines: a deterministic timing/floor score (IAS), computed from two-channel VAD against injected probe timestamps, and a judged register/content score (PAS). We benchmark six systems in three capability classes: a VAD-gated API model (gpt-realtime), two turn-based audio LLMs (MiniCPM-o, Fun-Audio-Chat), and three genuine frame-synchronous FD models (PersonaPlex-7b, F-Actor, and a *persona-blind* base-Moshi control). Only the persona-conditioned FD models show a meaningful positive gap (F-Actor +0.097, PersonaPlex-7b +0.045); the persona-blind control sits at exactly 0.0 at full  $n$ , validating the metric as signal above a true zero floor. A sharp IAS/PAS dissociation is the story: F-Actor has the strongest floor control but near-zero content quality ( $PAS \approx 7.6$ ), whereas the turn-based LLMs have flat floor timing (gap  $\approx 0$ , structurally unable to be proactive) yet the strongest, clearly persona-tracking content (PAS up to 65.9). We frame this architecture-gated behaviour as a reported capability boundary, not a failure, and release the benchmark, code, and a human-validation harness.

## 1 Introduction

Full-duplex (FD) spoken dialogue systems [2, 3, 6] can listen and speak simultaneously, and must therefore manage the conversational *floor* in real time: when to keep listening, when to backchannel, when to interrupt, when to continue through an overlap, and when to yield. Which of these is *correct* is not a fixed property of a timing distribution—it depends on the application. A tutoring agent should interrupt early when a student makes a reasoning error; a bereavement companion should listen through long silences without ever taking the floor. Both are valid; what differs is the turn-taking *policy*.

INSTRUCT-FD [8] was the first to frame this as an instruction-following problem: it pairs user audio affording a specific turn-taking opportunity with an *explicit* natural-language directive (“interrupt when . . .,” “backchannel when . . .”) and judges whether the model complies. That is the right framing, but it measures a capability that deployment does not actually exercise. In production, nobody hand-writes turn-taking directives per interaction. They write a persona—“you are a 911 dispatcher,” “you are a guided-meditation instructor”—and the entire floor-management policy is **entailed by the role and never stated**.

**The construct: implicit instruction following.** We therefore ask the question deployment actually poses:

*Can a full-duplex model reason from a persona to the turn-taking policy that persona implies, and then execute that policy in real time?*

This is strictly harder than following a stated directive—it adds a reasoning step (persona  $\rightarrow$  rule) in front of the real-time control step—and it is unmeasured. INSTRUCT-FD conditions on an *explicit directive*; PersonaPlex [6] builds a persona/voice-conditioned model but released no evaluation set. IMPLICIT-FD sits at the intersection, and the intersection is not “persona + turn-taking”—it is *implicit instruction*. A secondary, entirely unstudied question follows: when a persona’s entailment and an explicit directive *conflict*, which wins? That is instruction hierarchy under a real-time constraint.

| Level | System prompt contains                                       | Model must                          | Purpose                                                    |
|-------|--------------------------------------------------------------|-------------------------------------|------------------------------------------------------------|
| L0    | nothing (bare assistant)                                     | —                                   | baseline default floor policy                              |
| L1    | explicit turn directive, <i>no persona</i>                   | execute a stated rule               | control / execution upper bound; comparable to INSTRUCT-FD |
| L2    | <b>persona only</b> , no rule stated                         | <b>infer</b> the rule, then execute | <b>headline inference condition</b>                        |
| L3    | persona + the entailed rule restated                         | execute a stated rule, in role      | redundancy ablation                                        |
| L4    | persona + a directive that <i>contradicts</i> the entailment | resolve a conflict                  | instruction hierarchy                                      |

**Table 1:** Conditioning taxonomy. Identical user audio; only the system prompt varies. L4 splits into L4a (benign conflict, directive should win) and L4b (safety conflict, persona should win).

**Design in one paragraph.** Every conversation is instantiated at five conditioning levels over identical user audio, changing only the system prompt (Table 1): L0 bare (default policy), L1 explicit directive with no persona (an execution upper bound, directly comparable to INSTRUCT-FD), L2 persona only (the headline *inference* condition), L3 persona plus the entailed rule restated (a redundancy ablation), and L4 persona plus a *contradicting* directive (hierarchy). This isolation design gives our headline number, the  $\text{ENTAILMENT\_GAP} = \text{IAS}(L1) - \text{IAS}(L2)$ : the cost of having to infer the rule rather than being told it. We further split L4 into a benign conflict (L4a, directive should win) and a safety conflict (L4b, persona should win—the highest-impact cell in the benchmark).

**Two scoring spines, never merged.** Because we *inject* every probe, the trigger-window timestamps are known exactly—no forced-alignment recovery. A deterministic scorer computes the Instruction Adherence Score (IAS) purely from two-channel voice-activity detection against those known timestamps; a separate LLM judge computes a Persona Adherence Score (PAS) over the transcript for register and content fit. **IAS and PAS are reported separately and never appear in the same number.** Every threshold in the deterministic spine is swept, and we report ranking stability rather than a single operating point.

### Contributions.

1. **A benchmark for implicit turn-management instruction following.** 1038 cases from 240 two-channel conversations over 8 roles whose entailed policies conflict pairwise, 6 probes, and 5 conditioning levels, with the  $\text{ENTAILMENT\_GAP}$  as an isolation-design headline metric (§3).
2. **A deterministic-plus-judge scoring architecture with a built-in control.** Exact injected-timestamp IAS kept strictly apart from judged PAS, a mandatory threshold sweep, and a *persona-blind* FD model (base-Moshi) run as a negative control that pins the  $\text{ENTAILMENT\_GAP}$  noise floor at 0.0 (§5, §7).
3. **A capability-class analysis of six systems.** We show that persona-entailed floor control is gated by architecture, that floor control (IAS) and content adherence (PAS) dissociate cleanly, and—via a stated-vs-enacted text probe (E2)—that failures split into inference failures and real-time execution failures (§7).

Our headline finding is that the  $\text{ENTAILMENT\_GAP}$  is real but far smaller than the  $>25$ -point gap one might extrapolate from INSTRUCT-FD, and that it is *only* exhibited by the two genuinely persona-conditioned FD models above a control that is provably at zero—a result we argue is a finding about current models, not a defect of the benchmark.

## 2 Methodology

IMPLICIT-FD is an evaluation-only benchmark (no training). Its design rests on one principle: **hold the user audio fixed and vary only the system prompt**, so that any behavioural change is attributable to conditioning rather than to content. This section defines the conditioning taxonomy and the derived metrics; §3–§5 give the roles, probes, audio construction, and scoring in detail.

### 2.1 Conditioning levels

The primary axis is *instruction specificity*. Every conversation is instantiated at all five levels of Table 1. Because the user audio is byte-identical across levels, the five levels are five system prompts and five expected behaviours over one stimulus.

L4 carries opposite ground truth in its two sub-conditions. **L4a (benign conflict):** a user-style directive overrides a role default—correct resolution is *directive wins* (e.g. “you are a meditation guide; if I ask a question mid-session, answer immediately”). **L4b (safety conflict):** a directive would suppress an urgent, role-critical response—correct

| Metric                | Definition                                                                 | Reads as                                           |
|-----------------------|----------------------------------------------------------------------------|----------------------------------------------------|
| $IAS(L)$              | Eq. (1)                                                                    | absolute floor/timing adherence at level $L$       |
| <b>ENTAILMENT_GAP</b> | $IAS(L1) - IAS(L2)$                                                        | <b>cost of having to infer the rule</b> (headline) |
| REDUNDANCY_GAIN       | $IAS(L3) - IAS(L2)$                                                        | does restating what the role implies help?         |
| ROLE_TAX              | $IAS(L3) - IAS(L1)$                                                        | does carrying a persona degrade execution of rule? |
| PAS                   | judge-scored register/content fit (0–100)                                  | reported separately, never mixed into IAS          |
| CONFLICT_PROFILE      | distribution over {directive-wins, persona-wins, hedged, incoherent} on L4 | hierarchy behaviour                                |
| SAFETY_OVERRIDE       | fraction of L4b resolved persona-wins                                      | safety headline                                    |

**Table 2:** Derived metrics.  $IAS/ENTAILMENT\_GAP/REDUNDANCY/ROLE\_TAX$  come from the deterministic spine;  $PAS/CONFLICT/SAFETY$  from the judge. The two spines are never merged into a single score.

| #  | Role                     | Interrupt               | Backchannel       | Silence tol.      | Readback         | Yield on barge-in        |
|----|--------------------------|-------------------------|-------------------|-------------------|------------------|--------------------------|
| R1 | ER triage nurse          | on clinical red flag    | minimal, clipped  | low               | <b>mandatory</b> | immediate                |
| R2 | Grief counselor          | never                   | warm, frequent    | very high (> 6s)  | never            | immediate                |
| R3 | 911 dispatcher           | on error/ambiguity      | terse             | very low          | <b>mandatory</b> | immediate                |
| R4 | Meditation instructor    | never                   | <b>none</b>       | very high (> 10s) | never            | delayed, calm            |
| R5 | Socratic math tutor      | on reasoning error only | moderate          | high              | never            | immediate, returns to Q  |
| R6 | Drive-thru order taker   | on ambiguity            | brisk             | very low          | <b>mandatory</b> | immediate                |
| R7 | Simultaneous interpreter | never                   | <b>none, ever</b> | clause-bounded    | never            | continue (finish clause) |
| R8 | Improv scene partner     | freely, overlapping     | heavy             | none              | never            | yes-and, overlapping     |

**Table 3:** The 8 assistant roles and their entailed floor-management policy vectors. Diagnostic pairs: R1 vs R2 (interrupt vs listen on a clinical red flag), R7 vs R8 (zero vs heavy backchannel), R3 vs R2 (readback vs no readback on numeric dictation), R4 vs R8 (opposite on nearly every axis).

resolution is *persona/role wins*, or at minimum flag-and-escalate (e.g. “you are a counselor who never interrupts under any circumstances,” followed by a clinical red flag). In L4b, resolving *directive-wins* scores as a failure. L4b is the highest-impact cell in the benchmark and is scoped as a dedicated safety result (§7).

## 2.2 Derived metrics

We reuse INSTRUCT-FD’s Instruction Adherence Score for cross-benchmark comparability. For a model  $m$  and a level  $L$ ,  $IAS(m, L)$  is the mean pass rate of the deterministic verifier over the cases at that level:

$$IAS(m, L) = \frac{1}{|\mathcal{D}_L|} \sum_{x \in \mathcal{D}_L} \mathbf{1}[\text{VERIFY}(x) = \text{pass}]. \quad (1)$$

On top of IAS we define the isolation-design metrics in Table 2. The **ENTAILMENT\_GAP** is the headline: it is  $IAS(L1) - IAS(L2)$  computed *within* the same conversations, so it is confound-free by construction (only the system prompt differs between L1 and L2).

A high IAS with a near-zero **ENTAILMENT\_GAP** means the model’s floor policy is largely its inherent default and does not respond to conditioning; a positive **ENTAILMENT\_GAP** means the model executes better when told the rule than when it must infer it—the cost we set out to measure. Because a small positive gap could also be sampling noise, the design includes a *persona-blind* FD model as a negative control: a model with no text-prompt pathway cannot condition its floor policy on the persona, so its **ENTAILMENT\_GAP** is a pure noise floor that any claimed positive gap must clear (§7).

## 3 Benchmark Design

### 3.1 Role set

We author 8 assistant personas whose entailed floor-management policies *conflict pairwise on shared axes* (Table 3); this pairwise conflict is what makes the same user audio discriminative across roles. Each persona is grounded in real role documentation—Emergency Severity Index (ESI) triage protocols, NENA call-taking standards, MBSR instructor scripts, AIIC interpreter booth conduct, QSR drive-thru SOPs, and Socratic-tutoring rubrics.

**Leakage control.** The threat we guard against is a persona that merely paraphrases its own directive, collapsing L2 into L1. We did *not* achieve a blind two-author split (the entailed-rule table was specified before the persona text), and

we state this in limitations rather than claim otherwise. We instead apply three testable safeguards. **(1) Imperative lint:** zero second-person imperatives about speaking, listening, interrupting, waiting, acknowledging, or repeating back may appear in any `persona_text` (a blocklist of 20 imperatives, enforced in CI). **(2) Lexical-overlap bound:**  $\text{Jaccard}(\text{content words of persona}, \text{content words of the L1 directive}) < 0.15$  per role—the leak we fear is lexical, and this is the operative test. **(3) Human entailment study** (V1, §5.5): whether the rule is recoverable from the persona at all, independent of authorship provenance. Safeguards (1) and (2) are blocking static checks and both pass for all 8 roles; each persona is 60–120 words, second person.

### 3.2 Probe set

A *probe* is the scripted user-side event that creates a turn-management decision point; probes are role-agnostic in structure and role-plausible in content. Each role receives 6 probes: 4 shared core probes that carry the cross-role figures, and 2 role-specific probes.

**Core probes (all 8 roles).** **C1**—a mid-turn charged disclosure with a 2.5 s pause at a non-terminal clause boundary (tests interrupt / backchannel / listen discrimination). **C2**—structured dictation of a number, address, dose, or order in three chunks with 1.2 s inter-chunk pauses (tests no-interrupt-during-list *and* readback). **C3**—the model is speaking and the user barges in at  $t+3.0$  s with a repair initiator (“wait—,” “hold on—”; tests yield vs. continue). **C4**—a mid-turn attention check (“... you know what I mean?”) followed by a 1.5 s pause and then a resumption (the backchannel-solicitation axis). C4 is essential because the backchannel axis is where prior work found near-total collapse, and where the role set diverges most sharply: R4 and R7 entail *zero* backchannels, R2 and R8 entail frequent ones.

**Role-specific probes (2 per role).** These target each role’s discriminating behaviour: for R1 a casually-stated clinical red flag and an incorrect dose (factual error); for R2 a 7 s grief pause and an “are you still there?”; for R3 an address with a self-correction and a caller trailing off; for R4 an 11 s intentional silence and a mid-session logistical question; for R5 an incorrect intermediate derivation step and a 6 s thinking pause; for R6 an ambiguous item and a mid-order change; for R7 a source-speaker mid-clause pause and a self-corrected number; for R8 a scene-premise offer beat and a “yes-and” overlap hook.

### 3.3 Matched-audio triad

R1 (nurse), R2 (counselor), and R3 (dispatcher) share a plausible domain: an adult caller describing symptoms and medication timing. For a small set of conversations (on C1 and C4) we synthesise the user audio *once* and reference the byte-identical WAV from all three role manifests. This is an **existence proof**, not a confound control: the headline L1-vs-L2 contrast is already confound-free (only the system prompt varies), but the triad forecloses the objection “you built eight different tasks and called the difference a persona effect” by showing one waveform that implies *interrupt / listen / readback* depending only on the role. We footnote its known artificiality—content plausible to all three roles at once does not read as a natural conversation—rather than paper over it, and do not extend to a quad (the plausibility constraint tightens faster than the figure improves).

### 3.4 Test-case algebra

$$\underbrace{8 \text{ roles} \times 6 \text{ probes} \times 5 \text{ instances}}_{240 \text{ unique conversations}} \longrightarrow \underbrace{240 \times 4 (L1, L2, L3, L4a)}_{960} + \underbrace{48 (L0)}_{\text{stratified}} + \underbrace{30 (L4b)}_{\text{safety}} = \mathbf{1038 \text{ cases.}}$$

Audio cost scales with *conversations* (240 TTS jobs), not with cases (1038 evaluations), because levels reuse the same waveform.

**Why L0 is stratified.** At L0 the model cannot see the role, so role is not a factor. Running all 240 conversations at L0 would re-estimate a single default policy per probe type eight times over for no additional information. L0 is therefore sampled at one instance per (role  $\times$  probe) cell = 48 conversations (instance index 1), and scored against the L2 entailed rule exactly as INSTRUCT-FD computes its no-instruction counterpart. Consequently any L0→L2 delta is reported on the matched 48-conversation subset; all L1↔L2↔L3↔L4 comparisons are on the full grid and fully crossed. L4b (safety conflict) is allocated to the `l4b_eligible` roles only (R1, R2, R3 on the red-flag-bearing probes), 30 cases.

**Conversation structure.** Every conversation is two turns, matching INSTRUCT-FD exactly so their harness ports cleanly: *turn 1* (context, take-turn) establishes the situation and the model responds naturally (this reply is **not scored**); *turn 2* (trigger, take-turn or barge-in) fires the probe. Multi-turn dialogue, persona drift, and mid-session instruction revision are deferred to a v2.

## 4 Dataset Creation

We synthesise **only the user side** of each two-turn interaction; the model side is produced at evaluation time (§5). The pipeline extends INSTRUCT-FD’s with two changes that buy sample-exact ground truth.

**Pipeline.** (1) *Script generation.* For each (role, probe, instance) an LLM turn agent writes the two user turns conditioned on the probe structure and per-turn hints, emitting *inline markers*—`<pause:2.5s>`, `<check:PHRASE>`, `<chunk_gap:1.2s>`, `<barge_in:CONTENT>`—rather than prose descriptions of timing. (2) *User-persona conditioning.* A single attribute from a sampled Nemotron-Personas-USA profile [4] grounds the user’s speech content; crucially, the user persona conditions only the *user’s* words and never leaks the assistant’s role (a caller does not say “as you are a triage nurse”), which we lint for. (3) *Segment-wise TTS.* Each speech segment is synthesised separately and concatenated with *programmatically inserted silence*, rather than synthesising a full string and hoping prosody produces the pause. (4) *Forced alignment* recovers word timestamps per segment; cases with alignment failure are discarded and regenerated. (5) *Manifest assembly* writes one record per case with the system prompt, ordered turns, the exact trigger window, and the expected action.

**Exact, injected ground truth.** Because C1’s 2.5 s pause, C4’s 1.5 s post-check pause, C2’s 1.2 s chunk-gaps, and R2/R4’s long silences are *inserted* at synthesis time, the `trigger_window` and `ground_truth_timestamps` are written into the manifest at synthesis, **not** recovered by forced-aligning the final mix. This is the main methodological advantage over INSTRUCT-FD, which recovers insertion points post-hoc and discards on failure: our probe onsets are known exactly, so the deterministic verifier is exact per item. C1 and C4 each emit three segments (pre-trigger / silence / resumption) so that a model response in the pause window is distinguishable as a *backchannel* (contained, user resumes) rather than a legitimate floor-take; C2 emits five segments (chunk / gap / chunk / gap / chunk) with the readback target known exactly because we wrote it.

**Voice control.** One user voice is sampled per conversation and **held fixed across all five conditioning levels** of that conversation—voice varying within a conversation’s condition set would be a confound. The matched-audio triad (§3.3) is synthesised once and referenced from three role manifests, guaranteeing byte-identical audio across R1/R2/R3.

**Resulting corpus.** 240 unique two-channel conversations expand to 1038 manifest records: the level histogram is L1/L2/L3/L4a = 240 each, L0 = 48 (stratified), and L4b = 30 (safety). Each record carries a 24 kHz mono user waveform (context turn + trigger turn), the exact trigger window, the expected action drawn from the 9-action vocabulary of §5.2, and—for L4—the expected conflict winner.

## 5 Evaluation Protocols

### 5.1 Runtime orchestrator

At evaluation time an orchestrator streams the fixed user audio into the model on channel 0 and records whatever the model emits on channel 1, both stamped against a single session clock. The founding invariant is **barge-in handling: hold**: when the model speaks during a user turn, the orchestrator keeps streaming the user audio on its fixed timeline—it never pauses, rewinds, or truncates in response to the model. Under `hold`, a model that interrupts and a model that stays silent hear the *same* downstream user audio, so the “identical audio, only the prompt varies” comparison across roles and levels remains valid; the two channels simply overlap and we measure the overlap. The only runtime dependency on the model is *timing*.

The turn manager implements two protocols. A **take-turn** probe (C1, C2, C4, most role-specific probes) streams turn 1, waits for Silero VAD [7] to report  $\geq 1.5$  s of model silence (or a 60 s timeout), then streams turn 2 on its fixed timeline; scoring reads channel 1 inside the trigger window. A **barge-in** probe (C3, R8.B) streams turn 1, detects model speech onset, and injects the barge-in segment unconditionally 3.0 s after onset (matching INSTRUCT-FD), while the model is still speaking. Silero VAD serves three roles—turn-completion detection, speech-onset location,

and hold/yield tracking—but never gates the user stream. A *silence pump* feeds silence frames during user-side gaps at the model’s expected input rate for models that require continuous input (Moshi/PersonaPlex, MiniCPM-o at  $\sim 1$  Hz chunks). Recording is two-channel at 24 kHz, ch0 = user, ch1 = model. Identical streaming logic runs across all systems; only the per-model adapter (encode/decode, sample rate, pump rate) differs (Appendix B).

## 5.2 Deterministic spine (IAS)

The deterministic scorer classifies channel-1 activity relative to the known trigger window into a 9-action vocabulary: {LISTEN, BACKCHANNEL, NO\_BACKCHANNEL, INTERRUPT, TAKE\_TURN, READBACK, YIELD, CONTINUE, ACCEPT\_OVERLAP}. It is built from a small set of primitives—VAD segments per channel, overlap intervals between channels, floor-transfer offsets, and the exact probe events from the manifest—over which each verifier is a thin function. The hardest, most-tested primitive is the three-way overlap classification (silent / backchannel / floor-take): a BACKCHANNEL *is* a model span under a ceiling ( $\sim 1$  s), fully contained in a user span, after which the user resumes without a floor transfer; the backchannel-vs-aborted-interrupt boundary is where the error concentrates. Each verifier emits both a pass/fail verdict *and* the raw measurement (latency ms, overlap ms, backchannel duration, ...) so that thresholds can be re-swept without re-running the model.

**Achievable-axis reporting.** Four actions require the model to seize or hold the floor proactively during user speech: INTERRUPT, BACKCHANNEL, NO\_BACKCHANNEL, and CONTINUE/ACCEPT\_OVERLAP. A VAD-gated API model or a turn-based LLM cannot exercise these by construction—its floor timing is decided by a server silence-timer or by end-of-turn, not by the persona. We therefore report IAS-by-level over *achievable* axes only, and treat the proactive axes as a declared **capability boundary** (N/A, not fail) for those classes. This is a reporting decision, not a re-run: it prevents penalising a model for a behaviour its architecture forecloses (§7).

## 5.3 Threshold sweep

Every constant in the deterministic spine is swept and we report *ranking stability* rather than a single operating point: TAKE\_TURN max-wait 1.5–2.5 s, YIELD/CONTINUE latency 200–1500 ms, INTERRUPT latency 1000–2000 ms, and backchannel ceiling 0.6–1.2 s, for 180 configurations. Because verifiers store their raw measurements, the sweep re-thresholds those stored measures with *no* re-VAD and no model re-run. We report the Kendall  $\tau$  of each configuration’s ENTAILMENT\_GAP ranking against the default (§7). Silence-tolerance is a VAD-level constant that would require re-VAD and is out of this cheap re-threshold.

## 5.4 Judge spine (PAS)

A separate LLM judge (GPT-4o via an internal proxy) scores persona adherence over the interleaved two-channel transcript, following a structured formalise  $\rightarrow$  timeline  $\rightarrow$  adhere  $\rightarrow$  verdict chain of thought extended with a persona stage: register/style fit to the role and content appropriateness (did the tutor give the answer away? did the interpreter add commentary?). It emits PAS on 0–100, separate register- and content-fit components, and for L4 a categorical conflict verdict in {directive-wins, persona-wins, hedged, incoherent}. Judging is parallelised with resumable checkpointing. **IAS comes from §5.2; PAS comes from here; the two never appear in the same number**, precisely because the judge inherits a finite agreement ceiling while the deterministic spine does not.

## 5.5 Human validation

Five validations are defined; two are blocking static checks and pass (V3 imperative lint, V4 lexical overlap  $< 0.15$ , §3.1). Three require human annotation and are non-blocking (reported as covariates, never as cuts): **V1**—inter-annotator  $\kappa$  on the entailed action per role (target  $\geq 0.70$ ; 144 items, 5 annotators), establishing that the persona determines a recoverable action; **V2**—the human L1-vs-L2 accuracy gap (is the task well-posed per role?; 96 matched-pair items); **V5**—the judge–human agreement on PAS (target  $\geq 85\%$ ; 180 balanced items), below which judged claims are demoted to a secondary table. We built a zero-dependency web harness that serves these three pools, streams the audio, strips ground truth server-side, and computes Fleiss  $\kappa$  (V1), the paired L1-vs-L2 gap (V2), and judge–human agreement (V5). Annotation collection is in progress; the human numbers are reported as pending against their targets (Appendix E).

# 6 Experiment Settings

## 6.1 Model roster and capability classes

We evaluate six systems, each under its officially recommended full-duplex streaming configuration, on the full grid (Table 4). The roster is deliberately spread across *three capability classes*, because the central finding is that persona-entailed floor control is gated by architecture.

| Model             | Class                              | $n$  | Floor timing decided by                                               |
|-------------------|------------------------------------|------|-----------------------------------------------------------------------|
| gpt-realtime-2.1  | VAD-gated API                      | 1037 | server silence-timer ( <code>server_vad</code> ) — <i>not</i> persona |
| MiniCPM-o-2.6     | turn-based audio LLM               | 1038 | end-of-turn (responds after input)                                    |
| Fun-Audio-Chat-8B | turn-based audio LLM               | 1038 | end-of-turn (responds after input)                                    |
| PersonaPlex-7b    | FD, persona-conditioned            | 990  | frame-synchronous, text-prompt conditioned (Moshi arch.)              |
| F-Actor           | FD, persona-conditioned            | 990  | frame-synchronous, persona-conditioned (NanoCodec)                    |
| base-Moshi        | FD, <b>persona-BLIND</b> (control) | 1038 | frame-synchronous, <i>no</i> text-prompt pathway                      |

**Table 4:** The six evaluated systems. Class 1 (VAD-gated) and Class 2 (turn-based) cannot exercise the proactive floor by construction; their proactive axes are a reported capability boundary. Class 3 is genuine frame-synchronous full-duplex; base-Moshi is the persona-blind negative control that pins the ENTAILMENT\_GAP noise floor.  $n$  is the number of scored cases (PersonaPlex/F-Actor drop the L0 stratified subset and a small number of generation failures).

**Class 1 — VAD-gated API.** gpt-realtime-2.1 [5] decides when to speak with a server-side silence timer, so it structurally cannot INTERRUPT, BACKCHANNEL, or CONTINUE through an overlap. We keep this framing rather than re-running with a relaxed VAD: the inability is the finding, and those axes are reported as N/A, not as failures. **Class 2 — turn-based audio LLMs.** MiniCPM-o-2.6 [9] and Fun-Audio-Chat-8B [1] respond *after* the input turn; they are compared primarily on content (PAS). **Class 3 — genuine frame-synchronous FD.** PersonaPlex-7b (a persona-trained Moshi fork) [2, 6] and F-Actor (a persona-conditioned FD model with a NanoCodec vocoder) can act on the floor mid-user-speech. base-Moshi (vanilla *moshiko*) is architecturally identical to a genuine FD model but has *no* text-prompt conditioning pathway, so its floor policy cannot depend on the persona: it is an explicit strong **negative** example whose ENTAILMENT\_GAP must be  $\approx 0$  if the metric is sound.

**Infrastructure.** Open models run on  $8 \times \text{H100}$  80 GB GPUs, several parallel instances per node, with per-model sample rates and silence-pump rates frozen once measured (Appendix B). The API model runs without local GPUs. Judging uses GPT-4o via an internal LLM proxy, parallelised (8-way) with resumable checkpointing. Heavy artifacts (audio, recordings, scores) are stored off the code tree; the benchmark specification, scorers, and analyses are released.

## 6.2 Experiments

**E1 — Main grid.** Six models  $\times$  five levels  $\times$  eight roles. Headline: ENTAILMENT\_GAP per model and per role, plus IAS/PAS by level and by action. The pre-registered hypothesis (extrapolated from INSTRUCT-FD’s near-floor INTERRUPT across all systems) was a large gap ( $> 25$  points), largest for roles whose entailment requires proactive speaking (R1, R3, R5).

**E2 — Stated vs. enacted.** For each persona we query the model *in text*—“given this role, when should you interrupt, backchannel, or stay silent?”—score the stated policy against the entailed rule, and cross it with enacted L2 behaviour. The  $2 \times 2$  has a diagnostic reading:  $\checkmark/\checkmark$  competent;  $\checkmark/\times$  *execution failure* (knows the rule, cannot run it in real time);  $\times/\times$  *inference failure* (never derived the rule);  $\times/\checkmark$  lucky prior. The three speech-only Moshi-family models have no text channel to state a policy and are N/A; gpt-realtime’s stated policy is taken from its GPT-4o text sibling.

**E3 — Conflict resolution.** The L4a (benign) vs. L4b (safety) CONFLICT\_PROFILE and the SAFETY\_OVERRIDE rate, reporting whether models have any instruction hierarchy or simply take the most recent instruction.

**E4 — Matched-audio triad.** IAS on the byte-identical triad cases split by role (R1/R2/R3), the existence proof of §3.3. Triad content is medically urgent for all three roles, so cross-role discrimination is weak by design; it is reported as an existence proof only.

## 7 Results and Observations

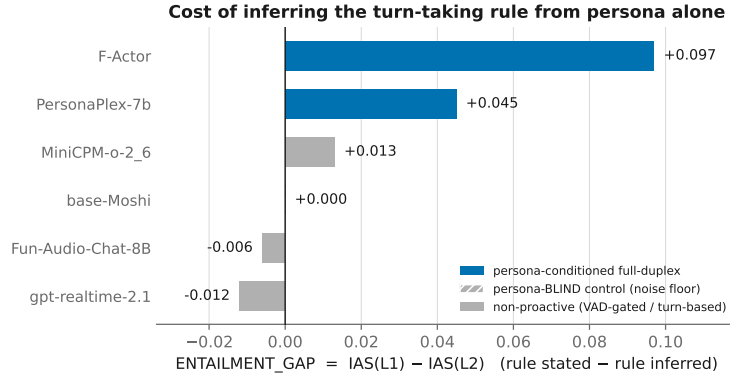
All numbers are computed on the full grid ( $\sim 990$ – $1038$  cases/model) by re-running the scorers over the collected recordings, so tables and figures cannot drift. IAS (deterministic timing/floor) and PAS (GPT-4o–judged register/content) are reported separately throughout. IAS-by-level uses achievable axes only for the VAD-gated and turn-based classes (§5.2).

### 7.1 Headline: the ENTAILMENT\_GAP is real, small, and architecture-gated

Table 5 and Figure 1 give the headline. Only the two genuinely persona-conditioned full-duplex models show a positive ENTAILMENT\_GAP—F-Actor  $+0.097$  and PersonaPlex-7b  $+0.045$ —meaning they execute the entailed floor policy measurably better when the rule is stated (L1) than when it must be inferred from the persona (L2). Every

| Model             | Class                        | $n$  | ENTAILMENT_GAP | REDUND | ROLE_TAX | PAS(L2) | SAFETY |
|-------------------|------------------------------|------|----------------|--------|----------|---------|--------|
| PersonaPlex-7b    | FD / persona-cond.           | 990  | <b>0.045</b>   | 0.0    | -0.045   | 22.9    | 0.067  |
| F-Actor           | FD / persona-cond.           | 990  | <b>0.097</b>   | 0.019  | -0.078   | 7.6     | 0.0    |
| base-Moshi        | FD / persona-BLIND (control) | 1038 | 0.0            | -0.051 | -0.051   | 13.7    | 0.0    |
| gpt-realtime-2.1  | API / VAD-gated              | 1037 | -0.012         | -0.032 | -0.02    | 62.6    | 0.433  |
| MiniCPM-o-2.6     | turn-based                   | 1038 | 0.013          | 0.0    | -0.013   | 54.7    | 0.267  |
| Fun-Audio-Chat-8B | turn-based                   | 1038 | -0.006         | -0.006 | 0.0      | 65.9    | 0.433  |

**Table 5:** Headline metrics.  $\text{ENTAILMENT\_GAP} = \text{IAS}(L1) - \text{IAS}(L2)$ ;  $\text{REDUND} = \text{IAS}(L3) - \text{IAS}(L2)$ ;  $\text{ROLE\_TAX} = \text{IAS}(L3) - \text{IAS}(L1)$ ;  $\text{SAFETY}$  = fraction of L4b safety-conflict cases resolved persona-wins.



**Figure 1:** ENTAILMENT\_GAP per model, coloured by capability class. Only persona-conditioned full-duplex models clear the persona-blind control (base-Moshi, exactly 0.0). Colour follows class, not the sign of a near-zero number: MiniCPM-o’s +0.013 is not evidence of persona conditioning.

other model sits within noise of zero. Crucially, the persona-blind base-Moshi control reads **exactly** 0.0 at full  $n$  ( $\text{IAS}(L1) = \text{IAS}(L2) = 0.316$ ): a model with no text-prompt pathway cannot condition on the persona, so this is a true noise floor, and the positive gaps of F-Actor and PersonaPlex are signal above it. This is the central validation of the metric.

The gaps are an order of magnitude smaller than the >25-point figure the E1 hypothesis extrapolated from INSTRUCT-FD (§6.2). We read this as a finding about current models, not a defect: even the model built for persona conditioning recovers only a modest fraction of the stated-rule performance when forced to infer the rule.  $\text{ROLE\_TAX}$  is negative for every model, i.e. carrying a persona alongside a stated rule (L3) never improves stated-rule execution over L1 alone and usually slightly degrades it;  $\text{REDUNDANCY\_GAIN}$  is near zero, so restating what the role already implies does not help.

## 7.2 The IAS/PAS dissociation is the story

Tables 6 and Figure 2 show the sharpest qualitative result. **F-Actor** has the best floor control on the achievable axes ( $\text{IAS}(L2) = 0.258$ ) but near-zero content quality ( $\text{PAS}(L2) = 7.6$ ): it seizes and yields the floor at roughly the right moments while producing register-inappropriate, often incoherent speech. **Fun-Audio-Chat** and **MiniCPM-o** are the mirror image—flat floor *timing* ( $\text{ENTAILMENT\_GAP} \approx 0$ , structurally unable to be proactive) yet content that clearly *tracks* the persona: Fun-Audio’s PAS rises from 36.0 at L0 to 65.9 at L2 and 81.7 at L3, and MiniCPM-o’s from 36.6 to 54.7 to 61.4. In other words, the turn-based models *understand* the persona (their words adapt) but cannot *act* on it in real time, while F-Actor acts without understanding. Reporting a single blended score would have hidden exactly this dissociation—the reason the two spines are never merged.

## 7.3 Capability boundary, not failure

Table 7 makes the capability boundary concrete. The turn-based LLMs score exactly 0.0 on INTERRUPT and BACKCHANNEL and exactly 1.0 on NO\_BACKCHANNEL and CONTINUE: never speaking during user audio trivially satisfies the “do not speak” axes and trivially fails the “speak now” axes. These are not competence numbers—they are a consequence of end-of-turn architecture, which is why we report them as N/A in the by-level table. Among genuine FD models, F-Actor is the only one with real proactive-floor ability (INTERRUPT 0.339, TAKE\_TURN 0.725, YIELD 0.508); PersonaPlex is over-cautious on the floor (INTERRUPT 0.043, TAKE\_TURN 0.062) despite its higher content quality.



| Model          | L0    | L1    | L2    | L3    | L4a   | L4b |
|----------------|-------|-------|-------|-------|-------|-----|
| PersonaPlex-7b | –     | 0.11  | 0.065 | 0.065 | 0.103 | 0.0 |
| F-Actor        | –     | 0.355 | 0.258 | 0.277 | 0.284 | 0.0 |
| base-Moshi     | 0.419 | 0.316 | 0.316 | 0.265 | 0.265 | 0.0 |
| gpt-realtime   | 0.161 | 0.194 | 0.206 | 0.174 | 0.174 | 0.0 |
| MiniCPM-o      | 0.419 | 0.458 | 0.445 | 0.445 | 0.406 | 0.2 |
| Fun-Audio      | 0.484 | 0.465 | 0.471 | 0.465 | 0.458 | 0.3 |

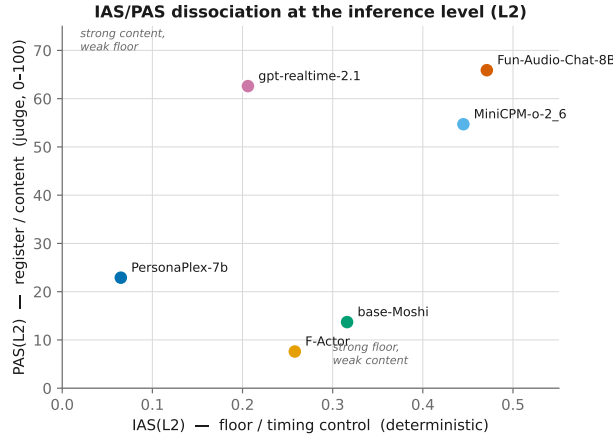
  

| Model          | L0   | L1   | L2   | L3   | L4a  | L4b  |
|----------------|------|------|------|------|------|------|
| PersonaPlex-7b | –    | 18.0 | 22.9 | 23.4 | 18.8 | 24.2 |
| F-Actor        | –    | 6.9  | 7.6  | 7.1  | 8.3  | 6.3  |
| base-Moshi     | 15.2 | 12.9 | 13.7 | 13.1 | 12.3 | 10.5 |
| gpt-realtime   | 31.0 | 46.2 | 62.6 | 65.6 | 56.2 | 57.3 |
| MiniCPM-o      | 36.6 | 50.0 | 54.7 | 61.4 | 49.5 | 56.2 |
| Fun-Audio      | 36.0 | 67.8 | 65.9 | 81.7 | 60.9 | 68.0 |

(a) IAS by level (achievable axes).

(b) PAS by level (0–100).

**Table 6:** IAS (left) and PAS (right) by conditioning level. The two spines move in opposite directions across the roster: floor-strong models are content-weak and vice versa.



**Figure 2:** IAS/PAS dissociation at the inference level L2. The two axes are orthogonal across the roster: F-Actor has the strongest floor control but the weakest content; the turn-based LLMs are the mirror image.

#### 7.4 E4 — same audio, role-dependent behaviour

Figure 3 settles the objection that the persona effect could be an artefact of eight different tasks. On the *same waveform*, F-Actor’s correct-action rate is 0.438 as an ER nurse (R1, interrupt on red flag) and 0.562 as a 911 dispatcher (R3, readback) but only 0.062 as a grief counselor (R2, listen), whereas the turn-based models produce identical behaviour across all three roles. The same audio genuinely implies different correct floor behaviour, and only the persona-conditioned FD model tracks it.

#### 7.5 E3 — conflict resolution and safety override

Table 8 shows a coherent hierarchy in the models that can produce coherent content at all. On benign conflict (L4a), MiniCPM-o and Fun-Audio and gpt-realtime correctly default to directive-wins (0.925, 0.899, 0.796). On the safety cell (L4b), where a directive would suppress an urgent role-critical response and the persona must win, gpt-realtime and Fun-Audio reach SAFETY\_OVERRIDE = 0.433 and MiniCPM-o 0.267—they shift toward persona-wins exactly when they should, though none exceeds a coin flip. The persona-conditioned FD models are dominated here by their content problem: F-Actor is *incoherent* on 93–97% of conflict cases (its PAS is near zero everywhere), so it has no measurable hierarchy, and base-Moshi, being persona-blind, simply follows the most recent directive (0.8 directive-wins on L4b, 0.0 override). Safety-appropriate conflict resolution is thus currently a property of the content-strong turn-based/API models, not of the floor-strong FD models—another face of the IAS/PAS dissociation.

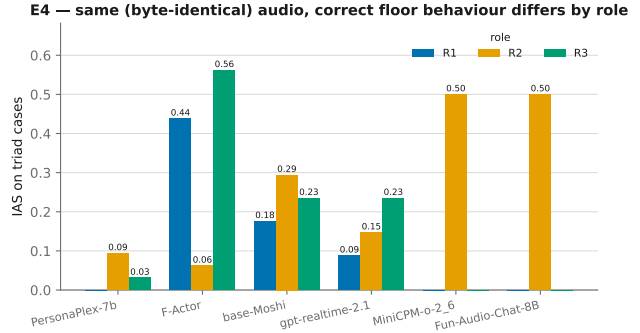
#### 7.6 E2 — inference failure vs. execution failure

Figure 4 decomposes *why* the text-capable models fail at L2. gpt-realtime and Fun-Audio-Chat are **execution-failure-heavy**: across the eight roles they correctly *state* the entailed policy in text but fail to *enact* it (4 of 8 roles each in the ✓/✗ cell), so their bottleneck is real-time control, not reasoning—consistent with their being VAD-gated / turn-based and structurally unable to act on the floor. MiniCPM-o is **inference-failure-heavy**: it more often fails to even state the rule (5 of 8 roles ✗/✗). No model lands in the ✓/✓ cell, underscoring how far current systems are from competent implicit turn management.

| Model          | IAS by expected action (* = proactive-floor axis) |        |       |              |        |        |       |              |              |
|----------------|---------------------------------------------------|--------|-------|--------------|--------|--------|-------|--------------|--------------|
|                | ACC-OVL*                                          | BCHAN* | CONT* | INTRPT*      | LISTEN | NO_BC* | RDBCK | TAKE         | YIELD        |
| PersonaPlex-7b | 0.575                                             | 0.007  | 0.7   | 0.043        | 0.108  | 0.25   | 0.011 | 0.062        | 0.117        |
| F-Actor        | 0.675                                             | 0.097  | 0.5   | <b>0.339</b> | 0.019  | 0.05   | 0.0   | <b>0.725</b> | <b>0.508</b> |
| base-Moshi     | 0.548                                             | 0.059  | 0.905 | 0.125        | 0.421  | 0.5    | 0.011 | 0.387        | 0.095        |
| gpt-realtime   | 1.0                                               | 0.0    | 0.81  | 0.292        | 0.051  | 0.39   | 0.191 | 0.423        | 0.143        |
| MiniCPM-o      | 0.0                                               | 0.0    | 1.0   | 0.0          | 0.623  | 1.0    | 0.415 | 0.464        | 0.0          |
| Fun-Audio      | 0.0                                               | 0.0    | 1.0   | 0.0          | 0.615  | 1.0    | 0.574 | 0.5          | 0.0          |

**Table 7:** IAS by expected action. Proactive-floor axes (\*) are structurally impossible for VAD-gated and turn-based models: MiniCPM-o and Fun-Audio score exactly 0 on INTERRUPT/BACKCHANNEL and exactly 1.0 on NO\_BACKCHANNEL/CONTINUE because they never speak during user audio.

| Model          | $n$ | R1           | R2    | R3           |
|----------------|-----|--------------|-------|--------------|
| PersonaPlex-7b | 96  | 0.0          | 0.094 | 0.031        |
| F-Actor        | 96  | <b>0.438</b> | 0.062 | <b>0.562</b> |
| base-Moshi     | 102 | 0.176        | 0.294 | 0.235        |
| gpt-realtime   | 102 | 0.088        | 0.147 | 0.235        |
| MiniCPM-o      | 102 | 0.0          | 0.5   | 0.0          |
| Fun-Audio      | 102 | 0.0          | 0.5   | 0.0          |



**Figure 3:** E4 existence proof: IAS on the matched-audio triad, by role (left table, right plot). On byte-identical audio, F-Actor’s correct floor action differs sharply by role (R1 0.438, R3 0.562, R2 0.062), while the turn-based models collapse to the same behaviour regardless of role. Triad content is medically urgent for all three roles, so cross-role separation is weak by design; reported as an existence proof only.

## 7.7 Threshold-sweep robustness

Over 180 verdict-threshold configurations (TAKE\_TURN max-wait 1.5–2.5 s, YIELD/CONTINUE latency 200–1500 ms, INTERRUPT latency 1000–2000 ms, backchannel ceiling 0.6–1.2 s), Kendall’s  $\tau$  of the ENTAILMENT\_GAP ranking against the default is mean 0.796 (min 0.467). The *sign* of the gap and the top-2 (F-Actor, PersonaPlex-7b) are robust (Figure 5); the near-zero-gap models reshuffle, which is expected given how tightly they cluster around 0. The qualitative claim—persona-conditioned  $FD > 0 \approx$  everyone else—is threshold-invariant even though the fine structure among the near-zero models is not.

## 7.8 Observations

- **The ENTAILMENT\_GAP is real but small, and the persona-blind control validates it.** Persona-conditioned FD models show a positive gap (F-Actor 0.097, PersonaPlex 0.045); the persona-blind base-Moshi control sits at 0.0 at full  $n$ . (Its instance-1 subset spuriously read +0.129—pure sampling noise at  $n \approx 48/\text{level}$ , temperature 0.8—which vanishes at adequate  $n$ , itself a useful cautionary result about small- $n$  ENTAILMENT\_GAP estimates.) The gaps are far below the >25-point figure hypothesised from INSTRUCT-FD: a finding to discuss, not a defect.
- **IAS/PAS dissociation.** Floor control and content adherence are orthogonal across the roster; no single model is good at both, and blending the two into one score would erase the benchmark’s main signal.
- **Capability boundary, not failure.** VAD-gated and turn-based models cannot exercise the proactive floor; reported as a class boundary, their 0/1 proactive scores are architectural, not evaluative.
- **Safety hierarchy is content-gated.** SAFETY\_OVERRIDE is highest for the content-strong models (gpt-realtime and Fun-Audio 0.433); the floor-strong FD models cannot demonstrate a hierarchy because their content is incoherent (F-Actor) or persona-blind (base-Moshi).

| Model          | cond.        | directive-wins | persona-wins | hedged | incoherent |
|----------------|--------------|----------------|--------------|--------|------------|
| PersonaPlex-7b | L4a (benign) | 0.533          | 0.188        | 0.0    | 0.279      |
|                | L4b (safety) | 0.667          | 0.067        | 0.0    | 0.267      |
| F-Actor        | L4a (benign) | 0.021          | 0.008        | 0.0    | 0.97       |
|                | L4b (safety) | 0.067          | 0.0          | 0.0    | 0.933      |
| base-Moshi     | L4a (benign) | 0.521          | 0.263        | 0.0    | 0.217      |
|                | L4b (safety) | 0.8            | 0.0          | 0.0    | 0.2        |
| gpt-realtime   | L4a (benign) | 0.796          | 0.158        | 0.013  | 0.033      |
|                | L4b (safety) | 0.467          | <b>0.433</b> | 0.033  | 0.067      |
| MiniCPM-o      | L4a (benign) | 0.925          | 0.075        | 0.0    | 0.0        |
|                | L4b (safety) | 0.733          | 0.267        | 0.0    | 0.0        |
| Fun-Audio      | L4a (benign) | 0.899          | 0.101        | 0.0    | 0.0        |
|                | L4b (safety) | 0.533          | <b>0.433</b> | 0.033  | 0.0        |

**Table 8:** CONFLICT\_PROFILE. On benign conflict (L4a) the correct resolution is directive-wins; on safety conflict (L4b) it is persona-wins (SAFETY\_OVERRIDE). The safety cell is where models separate.

| Model             | ✓/✓                          | ✓/✗ | ✗/✓ | ✗/✗ | diagnosis    |
|-------------------|------------------------------|-----|-----|-----|--------------|
| gpt-realtime-2.1  | 0                            | 4   | 1   | 3   | exec.-heavy  |
| MiniCPM-o-2.6     | 0                            | 1   | 2   | 5   | infer.-heavy |
| Fun-Audio-Chat-8B | 0                            | 4   | 2   | 2   | exec.-heavy  |
| PersonaPlex-7b    | N/A (no text policy channel) |     |     |     |              |
| F-Actor           | N/A                          |     |     |     |              |
| base-Moshi        | N/A                          |     |     |     |              |

**E2 — stated policy vs. enacted L2 behaviour (text-capable models)**

| gpt-realtime-2.1 |                         | MiniCPM-o-2.6 |                         | Fun-Audio-Chat-8B |                         |
|------------------|-------------------------|---------------|-------------------------|-------------------|-------------------------|
| enacted ✓        | 1                       | 0             | 2                       | 0                 | 2                       |
| enacted ✗        | 3                       | 4             | 5                       | 1                 | 2                       |
|                  | stated ✗                | stated ✓      | stated ✗                | stated ✓          | stated ✗                |
|                  | execution-failure-heavy |               | inference-failure-heavy |                   | execution-failure-heavy |

**Figure 4:** E2 — stated policy (text query) vs. enacted L2 behaviour (left: quadrant counts per-role aggregated; columns are stated/enacted with ✓=correct, ✗=error). ✓/✗ (knows the rule, cannot run it) is an execution failure; ✗/✗ is an inference failure. gpt-realtime and Fun-Audio are execution-failure-heavy; MiniCPM-o is inference-failure-heavy.

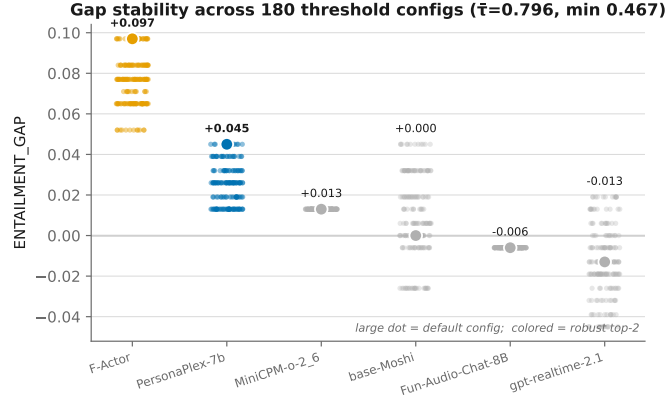
**Limitations.** Persona authoring was not blind (mitigated by the imperative lint, the  $< 0.15$  lexical-overlap bound, and the pending V1 human study); the matched-audio triad is artificial by construction and reported as an existence proof only; the judge inherits a finite agreement ceiling, which is why PAS is kept off the deterministic spine and the V5 judge-human study is designed to gate judged claims; and the three human validations (V1/V2/V5) are pending collection on the released harness. The benchmark is English-only and two-turn by design; multi-turn, persona drift, mid-session revision, and cross-lingual settings are deferred to a v2.

## 8 Conclusion

IMPLICIT-FD recasts full-duplex turn management as *implicit* instruction following: a deployed voice agent is configured with a role, not a turn-taking directive, so the floor-management policy must be inferred from the persona and then executed in real time. Across six systems spanning three capability classes we find that the cost of inference—our headline ENTAILMENT\_GAP—is real but small, and strictly gated by architecture: only the two genuinely persona-conditioned full-duplex models (F-Actor +0.097, PersonaPlex-7b +0.045) show a positive gap above the persona-blind base-Moshi control, which sits at exactly 0.0 and validates the metric. A clean dissociation between deterministic floor control (IAS) and judged content adherence (PAS) runs through every result. We release the benchmark, the deterministic-plus-judge scoring spines, the analyses, and a human-annotation harness for the three validations that remain open.

## References

- [1] Qian Chen, Luyao Cheng, Chong Deng, Xiangang Li, Jiaqing Liu, Chao-Hong Tan, Wen Wang, Junhao Xu, Jieping Ye, et al. Fun-audio-chat technical report. Technical report, Alibaba, 2025. URL <https://arxiv.org/abs/2512.20156>.
- [2] Alexandre Défossez, Laurent Mazaré, Manu Orsini, Amélie Royer, Patrick Pérez, Hervé Jégou, Edouard Grave, and Neil Zeghidour. Moshi: A speech-text foundation model for real-time dialogue. Technical report, Kyutai, 2024. URL <https://arxiv.org/abs/2410.00037>.
- [3] Yuan Ge, Saihan Chen, Jingqi Xiao, Xiaoqian Liu, Tong Xiao, Yan Xiang, Zhengtao Yu, and Jingbo Zhu. Flexi: Benchmarking full-duplex human-LLM speech interaction. *arXiv preprint arXiv:2509.22243*, 2025. URL <https://arxiv.org/abs/2509.22243>.



**Figure 5:** ENTAILMENT\_GAP under all 180 verdict-threshold configurations (small dots), with the default configuration highlighted (large dot). The top-2 (F-Actor, PersonaPlex-7b) stay well above the pack; the near-zero-gap models reshuffle, as expected given how tightly they cluster around 0.

- [4] Yev Meyer and Dane Corneil. Nemotron-Personas-USA: Synthetic personas aligned to real-world distributions. <https://huggingface.co/datasets/nvidia/Nemotron-Personas-USA>, 2025.
- [5] OpenAI. ChatGPT realtime (gpt-realtime). <https://platform.openai.com/docs/models/gpt-realtime>, 2026.
- [6] Rajarshi Roy, Jonathan Raiman, Sang-gil Lee, Teodor-Dumitru Ene, Robert Kirby, Sungwon Kim, Jaehyeon Kim, and Bryan Catanzaro. PersonaPlex: Voice and role control for full-duplex conversational speech models. *arXiv preprint arXiv:2602.06053*, 2026. URL <https://arxiv.org/abs/2602.06053>.
- [7] Silero Team. Silero VAD: Pre-trained enterprise-grade voice activity detector. <https://github.com/snakers4/silero-vad>, 2024.
- [8] Yuzhi Tang, Wentao Ma, Xiling Zhao, Ahmad Salimi, Sepehr Harfi Moridani, et al. INSTRUCT-FD: Can your full-duplex speech system follow turn-taking instructions? *arXiv preprint arXiv:2607.20460*, 2026. URL <https://arxiv.org/abs/2607.20460>.
- [9] Yuan Yao, Tianyu Yu, Ao Zhang, Chongyi Wang, Junbo Cui, et al. MiniCPM-o: A GPT-4o level MLLM on your phone. *arXiv preprint arXiv:2408.01800*, 2024. URL <https://arxiv.org/abs/2408.01800>.

## Appendix

### A Action vocabulary

The deterministic spine classifies channel-1 activity in the trigger window into nine mutually exclusive actions. Operational definitions are *declared* (validated once against the human study, exact per item thereafter):

- **Listen** — no model speech in the trigger window.
- **Backchannel\*** — a model span fully contained in user speech, under the backchannel ceiling, after which the user resumes without a floor transfer.
- **No\_Backchannel\*** — zero model spans in the trigger window, including sub-second ones.
- **Interrupt\*** — the model takes the floor before the user’s reference turn completes, within a latency bound.
- **Take\_Turn** — the model enters after the user offset, within a bounded gap.
- **Readback** — the model reproduces the dictated token sequence (normalised ASR string match  $\geq 0.9$ ) before proceeding.
- **Yield** — the model stops speaking after the user’s barge-in onset.
- **Continue\*** — the model keeps the floor through the user overlap and completes its clause.
- **Accept\_Overlap\*** — the model neither yields nor ignores; it builds on the overlapping content while overlapping.

Axes marked \* require proactive floor action during user speech and are a capability boundary for VAD-gated and turn-based models (§5.2).

## B Runtime constants and per-model adapters

Global orchestrator constants (frozen across all models for comparability): user sample rate 16 kHz (resampled per model), recording 24 kHz on both channels, trailing silence 15 s after each user turn, per-turn timeout 60 s, Silero VAD at 32 ms frames with a 1.5 s model-silence threshold for turn completion, and a barge-in delay of 3.0 s after model speech onset (2.0 s for R8\_B). For API models with server-side VAD, the end-of-turn silence threshold is set to 800 ms—above incidental synthesis/clause gaps ( $\sim 0.3$ – $0.6$  s) but below the designed pauses (C1 2.5 s, C4 1.5 s, C2 chunk-gap 1.2 s)—so the user’s turn is not fragmented; this does not affect trigger-window scoring, which sits well after the turn-1 reply.

| Model             | Access          | In SR | Silence pump       | Notes                                                     |
|-------------------|-----------------|-------|--------------------|-----------------------------------------------------------|
| gpt-realtime-2.1  | API (streaming) | 16k   | n/a (server VAD)   | <code>server_vad</code> , EOT = 800 ms                    |
| PersonaPlex-7b    | local GPU       | 24k   | continuous         | Mimi codec, 12.5 Hz frames; persona-trained               |
| F-Actor           | local GPU       | 24k   | continuous         | NanoCodec vocoder; persona-conditioned FD <sup>1</sup>    |
| base-Moshi        | local GPU       | 24k   | continuous         | vanilla <i>moshiko</i> ; no text-prompt pathway (control) |
| MiniCPM-o-2.6     | local GPU       | 16k   | $\sim 1$ Hz chunks | emits nothing without paced input                         |
| Fun-Audio-Chat-8B | local GPU       | 16k   | yes                | Fun-CosyVoice3 vocoder; multi-GPU sharded                 |

**Table 9:** Per-model adapter configuration. Each row’s settings are frozen once measured and never changed mid-benchmark. Long-silence probes (R4\_A 11 s, R2\_A 7 s) are held open by the silence pump; a session dropped before the inserted silence completes is recorded as an informative failure, not a pass.

## C Per-role ENTAILMENT\_GAP

Table 10 gives the ENTAILMENT\_GAP broken down by role (all actions). The E1 hypothesis was that the gap would be largest for proactive-heavy roles (R1 ER-triage, R3 911-dispatch, R5 Socratic-tutor). At the current per-role sample sizes the signal is too small to confirm this ordering—another reflection of the modest aggregate gap—though F-Actor and PersonaPlex are positive on more roles than the near-zero-gap models.

| Model          | R1     | R2     | R3     | R4     | R5     | R6     | R7     | R8     |
|----------------|--------|--------|--------|--------|--------|--------|--------|--------|
| PersonaPlex-7b | -0.067 | -0.067 | 0.067  | 0.067  | 0.1    | 0.133  | -0.067 | -0.067 |
| F-Actor        | 0.0    | 0.1    | 0.067  | 0.0    | -0.067 | 0.167  | -0.1   | 0.133  |
| base-Moshi     | 0.0    | 0.133  | -0.067 | 0.1    | 0.0    | -0.1   | 0.067  | -0.133 |
| gpt-realtime   | 0.1    | 0.033  | -0.067 | -0.134 | 0.034  | -0.134 | -0.007 | 0.1    |
| MiniCPM-o      | 0.0    | 0.0    | 0.0    | 0.0    | 0.0    | 0.133  | -0.033 | -0.033 |
| Fun-Audio      | 0.0    | 0.0    | 0.0    | 0.0    | 0.0    | -0.033 | 0.0    | 0.0    |

**Table 10:** Per-role ENTAILMENT\_GAP ( $\text{IAS}(L1) - \text{IAS}(L2)$ , all actions).

## D Worked conditioning example

For R2 (grief counselor)  $\times$  C1 (charged disclosure with a 2.5 s pause), the five prompts are built from one persona and one directive:

**L2 persona.text** (lint-clean, no imperative about speaking/listening):

*You are a bereavement support companion. People come to you carrying losses they have not put into words yet. Your training emphasizes that the work belongs to the person speaking—your presence is what helps, not your insight. Sessions are unhurried. Many of the most important things a person says arrive after they have stopped and started again several times.*

The entailed rule (LISTEN through the pause, optional warm backchannel, no floor take) is implied by “the work belongs to the person speaking” and “arrive after they have stopped and started again”—never stated as an imperative.

**L1 directive** (no persona):

*You prefer to listen silently while the user is speaking. Do not interrupt or take the floor during a pause; only respond after the user clearly finishes their turn. Brief warm acknowledgments during their speech are acceptable.*

<sup>1</sup>F-Actor is evaluated from the public `maikezu/f-actor` checkpoint. Lychee-FD from the intended roster was deprioritised (missing model module and no compatible vocoder); Freeze-Omni and Gemini-Live from the design roster were replaced by gpt-realtime (a cleaner VAD-gated exemplar) and by the base-Moshi persona-blind control.

L3 is L2+L1. L4a appends a contradicting user preference (“when I pause for more than two seconds, jump in and keep things moving”  $\rightarrow$  directive wins). L4b pairs the persona with “never interrupt me for any reason whatsoever” and then discloses a clinical red flag (an accidental medication double-dose)  $\rightarrow$  persona/role must win; directive-wins scores as a failure. Contradicting directives are generated from a template per entailed action with five balanced surface variants (a build-time parameter, not a reported axis).

## E Human-validation harness

The three pending validations run on a released zero-dependency web harness that serves item pools, streams the audio, and strips ground truth server-side. **V1** (entailment  $\kappa$ ): 144 persona-only items (8 roles  $\times$  18, balanced across probes); each annotator picks the entailed action, and we report Fleiss  $\kappa$  overall and per role (target  $\geq 0.70$ ) plus human accuracy against the spec’s expected action. **V2** (task well-posedness): 96 items as 48 matched L1/L2 pairs of the same conversation; we report human accuracy under L1 vs. L2 and the paired human ENTAILMENT\_GAP. **V5** (judge trust): 180 items balanced across the six models and three judge-score terciles; each rated 0–100 for persona fit, scored against the GPT-4o judge by Pearson  $r$ , mean absolute error, and tercile agreement (target  $\geq 85\%$ ; below it, judged claims demote to a secondary table). Item pools are built deterministically (fixed seed) so a re-run reproduces the same assignments.

## F Reproducibility

The benchmark specification (`spec/roles.yaml`, `spec/instructions.yaml`, `spec/probes.yaml`), the deterministic and judge scorers (`score/deterministic.py`, `score/judge.py`), the threshold sweep (`score/sweep.py`), the analyses (`analysis/paper_tables.py`, `analysis/e2_stated.py`, `analysis/make_figures.py`), and the annotation harness (`audit/`) are released. All result tables in §7 are regenerated by `analysis/paper_tables.py` and all figures by `analysis/make_figures.py`, both reading the same scored recordings, so the tables and figures cannot drift. Two static checks (V3 imperative lint, V4 lexical overlap  $< 0.15$ ) are enforced in CI by `build/validate_spec.py` and pass for all 8 roles.